

Rules and the maze

The maze is a fixed, published standard, and the rules of a run decide how you should play it. Read this before you write any code.

01

The maze

We run to the classic 16 x 16 standard of the UK Micromouse and Robotics Society (UKMARS), so every maze file, simulator and write-up from the sport applies directly. Nothing is bespoke to Dublin.

Grid

16 x 16 cells, 2880 mm square

Clear passage

168 mm between walls (walls are 12 mm thick)

Wall height

50 mm

Finish

White wall sides, red tops, matt black floor

Posts

12 x 12 mm at every lattice point, even where no wall is fitted

Start

A corner cell, walled on three sides

Goal

The 2 x 2 block at the centre, with one entrance



Start in a corner, goal at the centre. Walls shown are illustrative only.

Design to these numbers. Posts give you a 180 mm feature to correct distance against, but they also cause sensor spikes you must not read as walls. The goal is a set of four cells, not a coordinate. Calibrate your sensors under the lights you will race in.

02 What counts as a micromouse

There is no weight, cost or height limit. The rules protect one thing: the mouse solves the maze itself.

01 Fits inside a 25 cm square

Measured on the floor, at all times. You cannot start compact and unfold.

02 Autonomous and self-contained

No remote control, tethers or off-board computing. No human input once the run starts.

03 Stays on the floor and uses the openings

No flying, jumping or climbing walls.

04 Does not damage or mark the maze

Nothing may be deposited in the maze, so no trail to navigate by.

05 Does not shed parts

The mouse that finishes is the mouse that started.

The unwritten rule: the maze is unseen. You cannot hard-code a route or tune to a layout. Your code has to work on a maze handed to you cold.

03 How a run is timed and scored

- You get **ten minutes** in the maze (fewer if the entry list demands it) and up to **five runs**.
- A run is timed from leaving the start cell to reaching the goal. Time between runs is not run time, but it does use up your ten minutes.
- Your score is your **best single run**. A slow exploratory first run does not drag it down.
- You may not touch a running mouse unless a judge says so. A requested recovery **erases the mouse's memory of the maze**. Where a touch is allowed and the run continues, expect **three seconds plus a tenth of the run time** added.

What that means: explore, bank a safe finish, then escalate speed run by run.

One recorded 45-second run beats four spectacular crashes and no time on the board. See [S2 Race strategy](#).

04 The Dublin format

The maze is the classic standard, but the build window is not: **six hours, one identical kit per team, design, build, program and race on the day.**

- Every team gets the same kit: ESP32, encoded motors, motor driver, IMU, distance sensors, BMS, buck converter and battery.
- Judging is the **fastest verified run to the centre**, with further awards for design and reliability.
- Full-size and mini practice mazes are open all day. One mini maze is built so the fastest route cuts diagonally.

[Event details, kit list and registration →](#)

NEXT →

S2 Race strategy

Hack Club Build Hub — a practical teaching library for hackathons, clubs and community events. All backgrounds welcome.

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